

Normability of Higher Complex Sobolev Quasinorms

Automatic math-discovery packet

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Source Question

The source paper is Thai Duong Do and Duc-Bao Nguyen, *Higher complex Sobolev spaces on complex manifolds*, arXiv:2405.06385. In Remark 2.4, after defining the q -star quasinorm $\|\cdot\|_{*,q}$ on the higher complex Sobolev spaces W_q^* , the authors ask whether there is an equivalent norm. The relevant passage is:

It is interesting to know if we can build a norm in W_q^* equivalent to $\|\cdot\|_{*,q}$. When $q \geq 2$, since the mass of elements in the defining sequence can be chosen independently with each other [...] it seems to us that such a norm does not exist.

However, it is not true in the global setting because one cannot bound c_j by some constant times c_{j+1} (this is a different point in the global setting). It is interesting to know if we can build a norm in W_q^ equivalent to $\|\cdot\|_{*,q}$. When $q \geq 2$, since the mass of elements in the defining sequence can be chosen independently with each other (one can consider the 1-dimensional case to see why), it seems to us that such a norm does not exist.*

Remark 2.5. Note that when the domain Ω is nice (e.g., convex domain), we have $W_1^* = W^*$. This restriction is from the fact that we cannot solve the equation $dd^c\varphi = T$ in every domain. We also note that the spaces W_q^* is decreasing in q .

By using induction on q and the idea of [Vig07] Proposition 4], it is easy to prove that W_q^* has the following compactness property.

Figure 1: Remark 2.4 in the source paper, page 8.

Claim

The answer is positive. For every $q \geq 1$, in both the local and global settings of the source paper, W_q^* admits a norm equivalent to $\|\cdot\|_{*,q}$. More concretely, if

$$B = \{\varphi \in W_q^* : \|\varphi\|_{*,q} \leq 1\},$$

then $\text{co}(B)$ is bounded for $\|\cdot\|_{*,q}$. Hence the Minkowski functional of $\text{co}(B)$ is an equivalent norm.

Definitions

In the local setting, $\Omega \subset \mathbb{C}^k$ is a bounded domain. A defining sequence for $\varphi \in W_q^*(\Omega)$ is a sequence of psh functions $(\varphi_1, \dots, \varphi_q)$, with $\varphi_0 = \varphi$, such that

$$d\varphi_{j-1} \wedge d^c\varphi_{j-1} \leq dd^c\varphi_j, \quad j = 1, \dots, q,$$

and the masses $\|dd^c\varphi_j\|$ are finite. The quasinorm is

$$\|\varphi\|_{*,q} = \|\varphi\|_{L^2(\Omega)} + \inf \sum_{j=1}^q \|dd^c\varphi_j\|^{1/2^j}.$$

In the global setting, (X, ω) is a compact Kähler manifold. A defining sequence for $\varphi \in W_q^*(X)$ is $((c_1, \varphi_1), \dots, (c_q, \varphi_q))$, where $c_j \geq 0$ and φ_j are qps, such that

$$d\varphi_{j-1} \wedge d^c\varphi_{j-1} \leq c_j\omega + dd^c\varphi_j, \quad j = 1, \dots, q.$$

The quasinorm is

$$\|\varphi\|_{*,q} = \|\varphi\|_{L^2(X)} + \inf \sum_{j=1}^q c_j^{1/2^j}.$$

Proof Intuition

The possible obstruction in Remark 2.4 is that the different levels of a defining sequence are not controlled by each other in the global setting. But normability does not require such interlevel control. It is enough to show that the convex hull of the quasinorm unit ball remains bounded.

Inside the unit ball, every individual level is bounded: each $\|dd^c\varphi_j\|$, or each c_j , is at most one. If we take a convex combination of functions, we take the same convex combination of their defining sequences. Convex combinations preserve psh/qps functions, and the basic Jensen inequality

$$d\left(\sum_i \theta_i u_i\right) \wedge d^c\left(\sum_i \theta_i u_i\right) \leq \sum_i \theta_i du_i \wedge d^c u_i$$

preserves the defining inequalities level by level. Therefore each level of a convex combination still has mass or constant at most one. The resulting quasinorm bound is at most $q+1$, uniformly over the whole convex hull.

Levelwise Convexity Lemma

Lemma 1. *Let $\theta_i \geq 0$, $\sum_i \theta_i = 1$, and let $u_i \in W^{1,2}$ be real-valued. Then, in the sense of positive $(1,1)$ -currents,*

$$d\left(\sum_i \theta_i u_i\right) \wedge d^c\left(\sum_i \theta_i u_i\right) \leq \sum_i \theta_i du_i \wedge d^c u_i.$$

Proof. For smooth functions this is the pointwise Hilbert-space inequality

$$\left|\sum_i \theta_i \xi_i\right|^2 \leq \sum_i \theta_i |\xi_i|^2$$

applied to the complex gradients. The general $W^{1,2}$ case follows by the same pointwise inequality for the a.e. defined weak gradients, hence in the associated positive current sense. $\square$

Main Theorem

Theorem 1. *For every $q \geq 1$, the quasinormed spaces $W_q^*(\Omega)$ and $W_q^*(X)$ of the source paper are normable. More precisely, the Minkowski functional of the convex hull of the $\|\cdot\|_{*,q}$ -unit ball is a norm $\|\cdot\|_{\text{cvx},q}$ satisfying*

$$\|\varphi\|_{\text{cvx},q} \leq \|\varphi\|_{*,q} \leq (q+1)\|\varphi\|_{\text{cvx},q}.$$

Proof. Write $Q(\varphi) = \|\varphi\|_{*,q}$ and

$$B = \{\varphi : Q(\varphi) \leq 1\}.$$

Since Q is homogeneous, B is balanced. Hence $U := \text{co}(B)$ is convex, balanced, and absorbing. It remains to prove that U is Q -bounded.

We first treat the global setting. Let

$$\varphi = \sum_{i=1}^N \theta_i \varphi^{(i)}, \quad \theta_i \geq 0, \quad \sum_i \theta_i = 1, \quad Q(\varphi^{(i)}) \leq 1.$$

For each i , choose a defining sequence

$$((c_{i,1}, \varphi_{i,1}), \dots, (c_{i,q}, \varphi_{i,q}))$$

for $\varphi^{(i)}$ with

$$\|\varphi^{(i)}\|_{L^2} + \sum_{j=1}^q c_{i,j}^{1/2^j} \leq 1 + \varepsilon.$$

Define

$$\Phi_j = \sum_{i=1}^N \theta_i \varphi_{i,j}, \quad C_j = \sum_{i=1}^N \theta_i c_{i,j}.$$

Convex combinations of qpsh functions are qpsh. By Lemma 1 and the defining inequalities for the $\varphi^{(i)}$,

$$\begin{aligned} d\Phi_{j-1} \wedge d^c \Phi_{j-1} &\leq \sum_i \theta_i d\varphi_{i,j-1} \wedge d^c \varphi_{i,j-1} \\ &\leq \sum_i \theta_i (c_{i,j} \omega + dd^c \varphi_{i,j}) = C_j \omega + dd^c \Phi_j. \end{aligned}$$

Thus $((C_1, \Phi_1), \dots, (C_q, \Phi_q))$ is a defining sequence for φ . Moreover,

$$C_j \leq (1 + \varepsilon)^{2^j}$$

for every j , because $c_{i,j}^{1/2^j} \leq 1 + \varepsilon$. Also

$$\|\varphi\|_{L^2} \leq \sum_i \theta_i \|\varphi^{(i)}\|_{L^2} \leq 1 + \varepsilon.$$

Consequently

$$Q(\varphi) \leq (1 + \varepsilon) + \sum_{j=1}^q (1 + \varepsilon) = (q + 1)(1 + \varepsilon).$$

Letting $\varepsilon \downarrow 0$, $Q(\varphi) \leq q + 1$ for every $\varphi \in U$.

The local proof is identical. If $(\varphi_{i,1}, \dots, \varphi_{i,q})$ are local defining sequences, set $\Phi_j = \sum_i \theta_i \varphi_{i,j}$. Lemma 1 gives the defining inequalities, and positivity of the currents gives

$$\|dd^c \Phi_j\| \leq \sum_i \theta_i \|dd^c \varphi_{i,j}\| \leq (1 + \varepsilon)^{2^j}.$$

The same $q + 1$ bound follows.

Now define

$$\|\varphi\|_{\text{cvx},q} := \inf\{t > 0 : \varphi \in tU\}.$$

Because U is convex, balanced, and absorbing, this is a seminorm. The bound $Q \leq q + 1$ on U makes it positive definite: if $\|\varphi\|_{\text{cvx},q} = 0$, then $Q(\varphi) \leq (q + 1)t$ for every $t > 0$, so $Q(\varphi) = 0$ and $\varphi = 0$. Thus it is a norm.

Finally, since $B \subset U$, homogeneity gives

$$\|\varphi\|_{\text{cvx},q} \leq Q(\varphi).$$

Conversely, if $\varphi \in tU$, then $Q(\varphi) \leq (q+1)t$. Taking the infimum over t gives

$$Q(\varphi) \leq (q+1)\|\varphi\|_{\text{cvx},q}.$$

This proves the equivalence. □

Corollary 1. *With the norm $\|\cdot\|_{\text{cvx},q}$, the spaces $W_q^*(\Omega)$ and $W_q^*(X)$ are Banach spaces whenever they are complete for $\|\cdot\|_{*,q}$. In particular, this applies to the complete spaces in Proposition 2.7 of the source paper.*

Proof. Equivalent quasinorms/norms define the same Cauchy sequences. The source paper proves completeness for $\|\cdot\|_{*,q}$ in Proposition 2.7. □

Verification Notes

The proof uses only three ingredients already present in the source setup: homogeneity of the quasinorm, convexity of the classes of psh/qsh functions, and Jensen’s inequality for squared Hilbert norms of gradients. The script `code/verify_convex_bound.py` checks the finite-dimensional scalar inequality and the levelwise q -bound in random samples. It is a sanity check, not a substitute for the proof.

Limitations and Novelty Check

The constructed norm is abstract: it is the Minkowski functional of the convex hull of the quasinorm unit ball. It does not give a simple formula in terms of a last defining-sequence mass, and it does not contradict the source paper’s observation that the proposed norm $\|\cdot\|_{**,q}$ need not be equivalent to $\|\cdot\|_{*,q}$ in the global setting.

The run indexes were searched for arXiv:2405.06385, higher complex Sobolev spaces, W_q^* , q -star quasinorms, and normability. No prior packet was found. A bounded web search on July 3, 2026 for the exact title, the exact phrase “It is interesting to know if we can build a norm”, and W_q^* /complex Sobolev quasinorm terms located only the arXiv source paper and no later answer.

References

- T. D. Do and D.-B. Nguyen, *Higher complex Sobolev spaces on complex manifolds*, arXiv:2405.06385.
- S. Rolewicz, *Metric Linear Spaces*, PWN, 1985. Standard reference for normability criteria of quasi-normed linear spaces. The proof above is self-contained via the Minkowski functional.

Human Review Recommendation

Check the current inequality in Lemma 1 under the exact normalization of d^c used in the source paper. The normalization cancels on both sides, so no constant should appear, but this is the only convention sensitive point. Also verify whether the source authors intended the question only for the global spaces; the proof covers both global and local versions.